

SENG1050 Web Technologies

Lab 1

Part A*

1. List two major differences between LAN and WAN.
2. Briefly describe 4 basic network topologies. Cite the advantage and disadvantage of each topology.
3. What is the difference between a backbone link and a backbone network? How does this relate to the Internet Backbone?
4. Explain the main difference between circuit switching and packet switching. Which switching technique does the Internet use?
5. What is the difference between *forwarding* and *routing*?
6. Explain what happens when router overrun occurs?
7. What is a protocol? Why does the Internet need to use protocols?
8. What architecture does the Internet use?
9. What does TCP stand for? What is the main service that the TCP provides?
10. What does IP stand for? What is the main service that it provides?
11. What is an IP address? How many IP addresses can a router have?
12. How many bits does an IP address have?
13. Assuming that a data packet is going to be sent to a recipient with an IPv4 address of 201.22.32.10. How will this information be presented in the IP header of the data packet?
14. Why does the Internet need DNS?
15. What is the protocol for sending emails over the Internet?
16. What application layer protocols will need to be involved from after you type "newcastle.edu.au" in your browser and press return, to when the browser displays the webpage?
17. What is the Internet? Is the Internet actually the WWW? Why or why not?

Part B

1. Interpret the HTTP header of "<https://en.wikipedia.org/wiki/HTTP/3>" by using:

a. Command Prompt

i. Open the command prompt.

Windows: Click "Start" -> type "cmd" -> press "Enter"

Linux/Mac: Open any terminal app

ii. Enter the following on the command line:

```
curl -IL https://en.wikipedia.org/wiki/HTTP/3
```

b. Firefox, Chrome or Safari

i. Load the page "<https://en.wikipedia.org/wiki/HTTP/3>".

ii. Right click anywhere on the page, and select "**Inspect**" to open the Inspector

iii. Navigate to the **Network** tab, select the request that matches the URL you entered. Refresh the page if you cannot see it.

iv. Select "**Headers**".

Answer the following questions:

a) What is the version of HTTP used? (Hint: Right-click on a column name in the Inspector and enable the 'Protocol' column)

b) What type of HTTP request has been used to fetch this page?

c) What status code can you see? What does that mean?

2. Repeat the steps and answer the questions in 1 (above) for "<https://en.wikipedia.org/wiki/HTTP/5>". What are your observations?

3. Find your local (private) IP address by using the following two methods:

Windows

1) Open the **Settings** app. Then go to **Network & Internet** -> **Advanced network settings**. From there scroll down and click on **Hardware and connection properties**.

2) Type `ipconfig/all` on the command prompt

Mac

1) Open **Network** in **System Preferences**

2) Type `ipconfig getifaddr en0` on Terminal if you're connected to WiFi, or `ipconfig getifaddr en1` if you're using Ethernet.

Linux

Type `ip address show` on the command line.

Answer the following questions:

a) What is your local (private) IP address?

b) Take note of the field 'DNS Servers'. What do you think these IP addresses represent?

c) Look at the field 'Subnet Mask'. What do you think this field means?

4. Find your external (public) IP address by searching for "*what is my IP address*" on a search engine (verified on [google.com](https://www.google.com) and [duckduckgo.com](https://www.duckduckgo.com)).

5. In the command prompt, type "ping google.com". What is the IP address? What do you

think would be the result of entering this IP address into a browser's address bar? Try it, was the result what you expected?